

# Elyssa Hofgard

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## EDUCATION

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|--------------|--------------------------------------------------------------------------------------------------------|----------------|
| 2022–present | <b>PhD Electrical Engineering and Computer Science (EECS)</b><br>Massachusetts Institute of Technology | GPA: 4.90/5.00 |
| 2021–2022    | <b>M.S. Computational and Mathematical Engineering</b><br>Stanford University                          | GPA: 3.97/4.00 |
| 2017–2021    | <b>B.S. Physics with Honors, Minors: Mathematics and History</b><br>Stanford University                | GPA: 3.91/4.00 |

## RESEARCH INTERESTS

- Structural, topological, and symmetry-based descriptions of disordered systems
- Emergent order and multipolar order parameters in amorphous and magnetic systems
- Symmetry-preserving and equivariant machine learning methods for materials characterization and discovery

## RESEARCH & EMPLOYMENT EXPERIENCE

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 2022–present | <b>Massachusetts Institute of Technology</b><br><i>Graduate Research Assistant, Atomic Architects, Professor Tess Smidt</i> <ul style="list-style-type: none"><li>• Developing E(3)-equivariant neural network architectures for applications in materials science and physics</li><li>• Incorporating interpretable symmetry breaking parameters into E(3)-equivariant networks to model phase transitions</li><li>• Determining crystal structure and lattice parameters from powder X-ray diffraction data using invariant, symmetry-aware representations</li></ul>                               |
| 2024–present | <b>Lawrence Berkeley National Laboratory</b><br><i>Collaboration, Dr. Sinéad Griffin &amp; Dr. Dmitriy Morozov</i> <ul style="list-style-type: none"><li>• Completed two summer practicums through the DOE CSGF (2024, 2026), continued collaboration throughout the year</li><li>• Characterizing length scales of order in amorphous materials, modeling multipolar order parameters in magnetic materials using ML and DFT codes</li><li>• Developed an efficient implementation of structural spillage calculations to identify band inversion in non-crystalline topological materials</li></ul> |
| 2021–2022    | <b>Tompkins Laboratory</b><br><i>Research Assistant, IRIS-HEP</i> <ul style="list-style-type: none"><li>• Funded as a research assistant through Institute for Research and Innovation in Software for High Energy Physics (IRIS-HEP)</li><li>• Project focused on improving Kalman Filter algorithms used in particle track reconstruction</li></ul>                                                                                                                                                                                                                                                 |

- Summer 2021**      **Lawrence Livermore National Laboratory**  
*Data Science Summer Institute (DSSI)*
- Project focused on improving seasonal predictions of precipitation and temperature over the western US using deep learning
- 2020–2021**      **ATLAS Experiment, CERN**  
*Research Assistant, Professor Lauren Tompkins*
- 1 of 11 students selected nationwide for the International Research Experience for Students (IRES) at CERN funded through the National Science Foundation
  - Honors thesis explored the use of machine learning techniques to search for dark photons in ATLAS data, improved analysis sensitivity 10 times
  - Thesis awarded Firestone Medal for Excellence in Undergraduate Research
- 2019–2021**      **Molecular Imaging Instrumentation Laboratory**  
Stanford, CA  
*Research Assistant, Professor Craig Levin*
- Led an experiment to develop novel algorithms for analyzing dual-isotope positron emission tomography (PET) systems
- Summer 2019**      **Regeneron Genetics Center**  
Tarrytown, NY  
*Summer Intern, R&D Data Engineering Group*
- Led the development of a statistical method to distinguish gene level effects across phenotypes
- 2018–2019**      **LIGO Group**  
Stanford, CA  
*Research Assistant, Professor Brian Lantz*
- Designed a wind fence for seismic isolation at the LIGO Hanford Site

## HONORS & AWARDS

- 2024**      **QuARC, MIT QSEC Annual Conference**  
*Best Poster in Quantum Algorithms and Machine Learning Session.*
- 2023**      **Department of Energy Computational Science Graduate Fellowship**  
*Accepted, up to 4 years of funding*
- 2023**      **National Defense Science and Engineering Graduate Fellowship**  
*Declined for DOE CSGF*
- 2022**      **MIT Advanced Television and Signal Processing Fellowship**
- 2022**      **NSF GRFP Honorable Mention**
- 2021**      **Firestone Medal for Excellence in Undergraduate Research**  
*Awarded for thesis with ATLAS experiment*
- Given to the top ten percent of all honors theses in the social sciences, natural sciences, and engineering and applied sciences at Stanford

2019

**American Physical Society Far West Section Meeting**  
*Honorable Mention for Best Undergraduate Poster*

## PUBLICATIONS

### PREPRINTS

- [4] R. McConkey, J. Balla, **E. Hofgard**, T. Smidt, A. Bodner, “Rotational equivariance and locality in data-driven subgrid-scale closures,” arXiv, 2026. Link: <https://arxiv.org/abs/2607.26850>. Under review.
- [3] **E. Hofgard**, K. Min, N. Segal, D.W. Mittan-Moreau, A. Mansouri Tehrani, V. Oklejas, J. Nigam, D.W. Paley, A.S. Brewster, T. Smidt, “Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants,” arXiv, 2026. Link: <https://arxiv.org/abs/2607.21829>. Under review.
- [2] R. McConkey, J. Balla, J. Bailey, A. Backour, **E. Hofgard**, T. Jaakkola, A. Bodner, T. Smidt, “Turbulence teaches equivariance to neural networks,” arXiv, 2026. Link: <https://arxiv.org/abs/2602.04695>. Under review.
- [1] R.B. Garg, **E. Hofgard**, L. Tompkins, and H. Gray, “Exploration of Different Parameter Optimization Algorithms within the Context of ACTS Software Framework,” arXiv, 2022. Link: <https://doi.org/10.48550/arXiv.2211.00764>.

### PEER-REVIEWED JOURNAL ARTICLES

- [3] **E. Hofgard**, A. Mansouri Tehrani, Y. Lin, S. Ji, T. Smidt, “Materials Characterization” (Section 7.4, pp 581–586), in “Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems,” *Foundations and Trends in Machine Learning*, 2025, Vol. 18: No. 4, pp 385–912. Link: <https://doi.org/10.1561/2200000115>.
- [2] The ATLAS collaboration, “Search for dark photons from Higgs boson decays via ZH production with a photon plus missing transverse momentum signature from pp collisions at  $\sqrt{s} = 13$  TeV with the ATLAS detector,” *J. High Energ. Phys.* 2023, 133. Link: [https://doi.org/10.1007/JHEP07\(2023\)133](https://doi.org/10.1007/JHEP07(2023)133). *Work recognized by the 2025 Breakthrough Prize in Fundamental Physics.*
- [1] D. Cotter, **E. Hofgard**, J. Novembre, A. Szpiech, and N. Rosenberg, “A Rarefaction Approach for Measuring Population Differences in Rare and Common Variation,” *Genetics*, 224(2), iyad070 (2023). Link: <https://doi.org/10.1093/genetics/iyad070>.

### COLLABORATION PUBLICATIONS

As a member of the LIGO Scientific Collaboration (2018–2019), co-author on 11 LSC publications on gravitational wave observations and detector characterization; full list available via Google Scholar.

### CONFERENCE PAPERS AND PROCEEDINGS

- [3] **E. Hofgard\***, H. Lawrence\*, V. Portilheiro, Y. Chen, T. Smidt, R. Walters, “To Augment or Not to Augment? Diagnosing Distributional Symmetry Breaking,” *International Conference on Learning Representations*, 2026. Link: <https://arxiv.org/abs/2510.01349>. \*denotes equal contribution.
- [2] R. Wang, **E. Hofgard**, H. Gao, R. Walters, and T. Smidt, “Discovering Symmetry Breaking in Physical Systems with Relaxed Group Convolution,” *Forty-first International Conference on Machine Learning*, February 2024. Link: <https://arxiv.org/abs/2310.02299>.

- [1] **E. Hofgard**, G. Chinn, and C. Levin, “Simultaneous Multi-Isotope PET: A Computational Framework for Line of Response (LOR) Identification,” *2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, 2020. Link: <https://doi.org/10.1109/NSS/MIC42677.2020.9507891>.

## WORKSHOP PAPERS

- [2] J. Balla\*, J. Bailey\*, A. Backour, **E. Hofgard**, T. Jaakkola, T. Smidt<sup>†</sup>, R. McConkey<sup>†</sup>, “Implicit Augmentation from Distributional Symmetry in Turbulence Super-Resolution,” *Machine Learning and the Physical Sciences Workshop, NeurIPS*, San Diego, USA, 2025. Link: <https://arxiv.org/abs/2509.20683>. \*, <sup>†</sup> denote equal contribution.
- [1] **E. Hofgard**, R. Wang, R. Walters, and T. Smidt, “Relaxed Equivariant Graph Neural Networks,” *Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM) at the Forty-first International Conference on Machine Learning*, July 2024. Link: <https://arxiv.org/abs/2407.20471>.

## CONFERENCES & TALKS

### INVITED TALKS

- [2] **E. Hofgard**. “Using E(3)-Equivariant Neural Networks (ENNs) to Uncover Symmetry-Implied Missing Information.” Computational Research in Boston and Beyond (CRIBB) Seminar, March 2025.
- [1] **E. Hofgard**. “Applications of Euclidean Neural Networks for the Design and Understanding of Physical Systems.” Plenary talk at The 5th International Conference on Data-Driven Plasma Science, Berkeley, CA. Given on behalf of Tess Smidt.

### CONFERENCE PRESENTATIONS AND POSTERS

- [7] **E. Hofgard**, G. Moore, O. Ashour, E. Banyas, T. Smidt, S.M. Griffin. “Using Equivariant Neural Networks to Learn Multipolar Order Parameters.” Oral Presentation at APS Global Physics Summit 2026.
- [6] **E. Hofgard**, D. Morozov, M.A. Hussien, T. Bayaraa, S.M. Griffin, T.E. Smidt. “Applications of Topological Data Analysis (TDA) for Studying Structural Motifs in Amorphous Bi<sub>2</sub>Se<sub>3</sub>.” Oral Presentation at APS Global Physics Summit 2025.
- [5] **E. Hofgard**, A.M. Tehrani, and T. Smidt. “Learning Crystal Structure from Powder X-Ray Diffraction Data Using Invariants.” Oral Presentation at 2024 American Physical Society March Meeting.
- [4] **E. Hofgard**, R. Wang, and T. Smidt. “Using Symmetry-Equivariant Neural Networks (ENNs) to Uncover Symmetry-Implied Missing Information.” Oral Presentation at 2023 Materials Research Society Fall Meeting.
- [3] **E. Hofgard** et al. “Exploration of Machine Learning (ML) Methods for Determining Unit Cell Parameters from Powder XRD Patterns.” Oral Presentation at 2023 Materials Research Society Fall Meeting.
- [2] **E. Hofgard**, and L. Tompkins. “Cuts Optimization and Machine Learning Models for Dark Photon Signal-Background Discrimination with the ATLAS Detector.” Oral presentation at 2021 APS April Meeting.
- [1] **E. Hofgard**. “An Algorithmic Approach to Identify Non-Pure Positron Emitters for Positron Emission Tomography (PET) Imaging Applications.” Poster presented at 2019 Annual Meeting of the APS Far West Section.

## SOFTWARE

**Structural Spillage Package:** Sole developer of a more efficient implementation of structural spillage calculations for identifying band inversion in non-crystalline topological materials, extending Griffin *et al.*, *Phys. Rev. Research* 5, L042011. <https://github.com/Griffin-Group/Structural-Spillage-Package>.

## TEACHING & ADVISING

Fall 2026                      Teaching Assistant, MIT 18.C06 (Linear Algebra and Optimization)

2024–present                Mentor, MIT Undergraduate Research Opportunities Program

- Supervising Kyuchool Min, an undergraduate MIT student, on representation learning for XRD data, resulting in co-authorship on Hofgard, Min, et al. “Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants.”
- Min submitted first-author work, “FrOGS: Discrete Neural Sampler for Independent Alloy Configurations Across Chemical Conditions,” to the NeurIPS AI4Mat Workshop (2026)

Fall 2020                      Teaching Assistant, Stanford Physics 21: Mechanics, Fluids, and Heat

Winter 2020                  Learning Assistant, Stanford Physics 41E: Mechanics

2019–2021                   Tutor, Schwab Learning Center, Stanford University

- Provided individual tutoring for students with learning differences; tutored linear algebra, multivariable calculus, and differential equations

## PROFESSIONAL SERVICE

### REVIEWING

- International Conference on Learning Representations (ICLR), 2026
- Conference on Neural Information Processing Systems (NeurIPS), 2026
- Geometry-grounded Representation Learning and Generative Modeling (GRaM) Workshop at ICML/ICLR, 2024, 2026
- Machine Learning and the Physical Sciences (ML4PS) Workshop at NeurIPS, 2024-2025

### LEADERSHIP

2024–2025                      Committee Member, IAIFI Equity and Inclusion Committee

*NSF AI Institute for Artificial Intelligence and Fundamental Interactions*

- One of two graduate student members on the committee to advise IAIFI on supporting early career members

2023–2024                      Co-President of Graduate Women in Course 6 (GW6)

- Elected as co-president of GW6, an organization to support graduate women in EECS at MIT

2022–2023                      MIT EECS Graduate Student Council (GSC) Representative

- Elected to advocate for the EECS student body

**2020–2021**      **Undergraduate Representative, Physics Undergraduate Studies Committee**

- Selected as one of two undergraduate representatives to offer ideas on course design and department culture

**2018–2020**      **Leadership, Physics Undergraduate Women and Gender Minorities at Stanford**

- Co-President, 2019–2020; Director of Community Development, 2018–2019

## **OUTREACH**

**2026**      **Invited Panelist, MIT EECS New PhD Cohort Orientation**

- Orientation event welcoming the incoming 2026 PhD cohort (113 students)

**2025–present**      **Mentor, MIT EECS Buddy Program**

- Mentoring and advising first-year PhD students on the transition to the PhD

**2024–2025**      **Invited Representative, MIT EECS Recruitment & Outreach**

- Panelist, MIT EECS Graduate Admissions Outreach, sharing the graduate student experience with prospective PhD students (2025)
- Selected by the EECS department to represent MIT at the Society of Women in Engineering (SWE) Conference (2024)